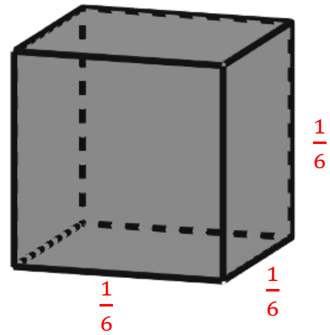
Grade 6 Accelerated　　Name_Answer Key________
Unit 8　　Date____________________
Lesson 1 Practice　　Period__________________

A $\frac{1}{6}$ inch unit cube is shown.

1. Determine the volume of the unit cube. Include the units for the cube's volume.

$V = lwh$

$V = \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{216}$ in^3

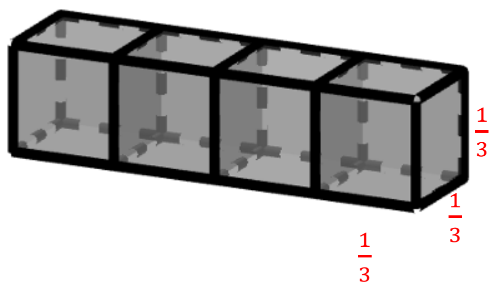


The diagram shows four unit cubes whose dimensions are $\frac{1}{3}$ inches.

2. The volume of one of the cubes is

___$\frac{1}{27}$___ cubic inches (in^3).

$\frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{27}$ in^3



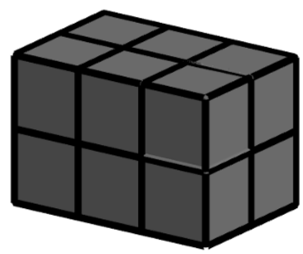
3. The volume of all four cubes is ___$\frac{4}{27}$___ in^3.

$4 \cdot \frac{1}{27} = \frac{4}{27}$

A rectangular prism is shown.

4. How many unit cubes whose dimensions are $\frac{1}{2}$ inch fill up the rectangular prism?

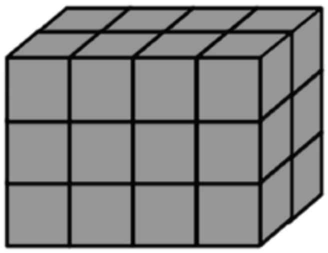
$(2)(3)(2) = 12$ cubes



5. Each unit cube whose dimensions are $\frac{1}{2}$ inch has a volume of $\frac{1}{8}$ cubic inches. Find the total volume, in cubic inches, represented in the diagram by multiplying the number of cubes that fill the rectangular prism by the volume of one of the cubes.

$12 \cdot \frac{1}{8} = \frac{12}{8} = 1\frac{1}{2}$ in^3

6. Each unit cube in the diagram has dimensions of $\frac{1}{4}$ inch. Find the total volume represented in the diagram, in cubic inches, using the formula, $V = lwh$.

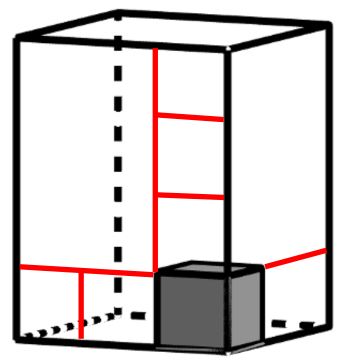


$\left(\frac{1}{4}\right)^3 = \frac{1}{64}$ in^3 per cube

$(4)(2)(3) = 24$ cubes

$V = \frac{1}{64} \cdot 24 = \frac{24}{64} = \frac{3}{8}$ in^3

A rectangular prism with one cube inside is shown.



7. How many cubes will it take to fill the rectangular prism?

$(3)(2)(4) = 24$ cubes

8. The rectangular prism has a length of $\frac{9}{4}$ centimeters (cm), a width of $\frac{6}{4}$ cm, and a height of 3 cm as shown. Determine the dimensions of the fractional unit cube.

Each unit cube has dimensions of $\frac{3}{4}$ in.

9. Determine the volume of the unit cube.

$\left(\frac{3}{4}\right)^3 = \frac{27}{64}$ in^3

10. Determine the volume of the rectangular prism.

$V = \frac{27}{64} \cdot 24 = \frac{648}{64} = 10.125$ in^3

OR

$V = \left(\frac{9}{4}\right)\left(\frac{6}{4}\right)(3) = 10.125$ in^3